Pronoun Incorporation and Resumption in Matsigenka

AKA David Inman wrote a generals paper

Language, Speakers

- ~6,000 speakers
- Arawakan





Methods

- Problems
  - Matsigenka is spoken in the remote Amazon
  - Difficult to access speakers
- Resources
  - Extensive translated corpus with hand-annotated IGT
  - Comprehensive dictionary
  - Good grammar sketch (for Nanti)
  - thanks to Lev Michael et al

Methods

- Corpus Study
  - 170 texts
  - >500 sentences
  - >36000 words
  - 4 stories reviewed manually to find instances of pronoun incorporation
  - IGT annotation allows searching for additional examples
  - Corpus studies take a while

Methods

- Counting clauses by hand gives idea of frequency of constructions
  - Without a parser, it's necessary to classify by hand

	Free Subject				Free Object		Both Free		
	S (s-)V	S V	(s-)V S	V S	(s-)V(-o) O	(s-)V O	S (s-)V(-o) O	S V(-o) O	(s-)V(-o) S O
	17	1	23	9	29	17	1	1	1
Total	50				46		3		

  - Two free arguments is hard to find
  - Both can occur with or without the agreement marker
  - Distribution of word orders gives hints for IS

Pronoun Incorporation

- Matsigenka verbs can take agreement marking for both subject and object.

  nokanakeri

  no-ok-an-ak-e-ri

  1S-leave-ABL-PERF-REALIS-3mO

  ‘I left him’

Pronoun Incorporation

– It's long been noticed that for many languages these "agreement markers" can also function as full pronouns (meaning there is no overt noun - *and the marker can anaphorically agree with a coreferent outside the VP*)

Pronoun Incorporation

- The most consistent test for PI vs agreement is non-local topic coreference

yogari itovaire onperaigakaro oshintoegi okajaigue**ri**ra oaku

y-oga=ri i-tovai-re o-onpera-ig-ak-a=ro o-shinto-egi o-ok-ja-ig-u-e=**ri**=ra oaku

3mS-that=CNTR 3mP-other-POSS 3fS-order-PL-PERF-REALIS=3fO 3fP-daughter-PL 3fS-throw-CL:fluid-PL-RET-IRREALIS=**3mO**=SUB river

Those others[masc], she ordered her daughters to throw **them[masc]** into the river.

  - Anaphoric (non-local) agreement of PI with external “those ohers”
  - The 3mO marker is therefore satisfying the (local) argument of its verb

Reducing Phenomena

- Other formalizations of this phenomenon end up introducing homophonous morphemes (Bresnan & Mchombo, LFG) or invisible pronouns (MP, some HPSG).
  - Neither of these are very satisfactory
  - We actually already have the machinery for PI
  - PI can be captured as a special form of morphologically-coded argument optionality

Reducing Phenomena

- Saleem (2010) introduces argument optionality into the Grammar Matrix
  - Items on VAL are marked as optional with OPT
  - OPT + features can be canceled via *head-opt-subj-phrase*, *head-opt-comp-phrase*
  - Capture PI: agreement marker constrains PNG values (allowing for anaphoric agreement), does not remove argument from valence list, marks itself as OPT +
  - One lexical entry per morpheme, reused PSRs

Reducing Phenomena

$$\begin{bmatrix} \textit{head-opt-subj-phrase} \\ \textbf{SYNSEM...SUBJ} \quad \langle\,\rangle \\ \textbf{HD-DTR...SUBJ} \quad \left\langle \begin{bmatrix} \textbf{OPT} & + \end{bmatrix} \right\rangle \end{bmatrix}$$

$$\begin{bmatrix} \textit{verb-subj-pi} \\ \text{SYNSEM...SUBJ} \quad \left\langle \begin{bmatrix} \text{OPT} & + \\ \boxed{1} & \end{bmatrix} \right\rangle \\ \text{HD-DTR...SUBJ} \quad \left\langle \begin{bmatrix} \text{OPT} & - \\ \boxed{1} & \end{bmatrix} \right\rangle \end{bmatrix}$$

Resumption

- Arguments of a verb may be realized twice: Once locally (incorporated pronoun, free pronoun, or NP), and once externally
- All external information structure *must be* coreferenced with a verb's argument

Resumption

- Contrastive focus

Ogari ivatsa yogavisaigavake**ro** kamatitya

o-oga=ri i-vatsa y-og-avis-ig-av-ak-e=**ro** kamatitya

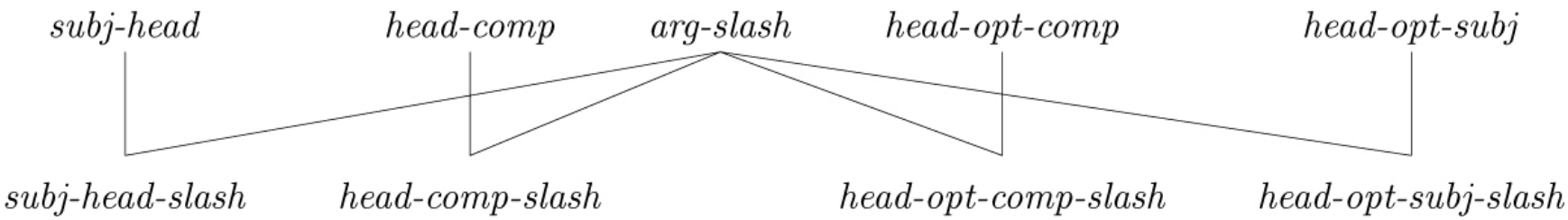
3fS-that=CNTR 3mP-body 3mS-CAUS-pass.by-PL-TRNS-PERF-REALIS=**3fO** downriver

His body, they made **it** pass downriver.

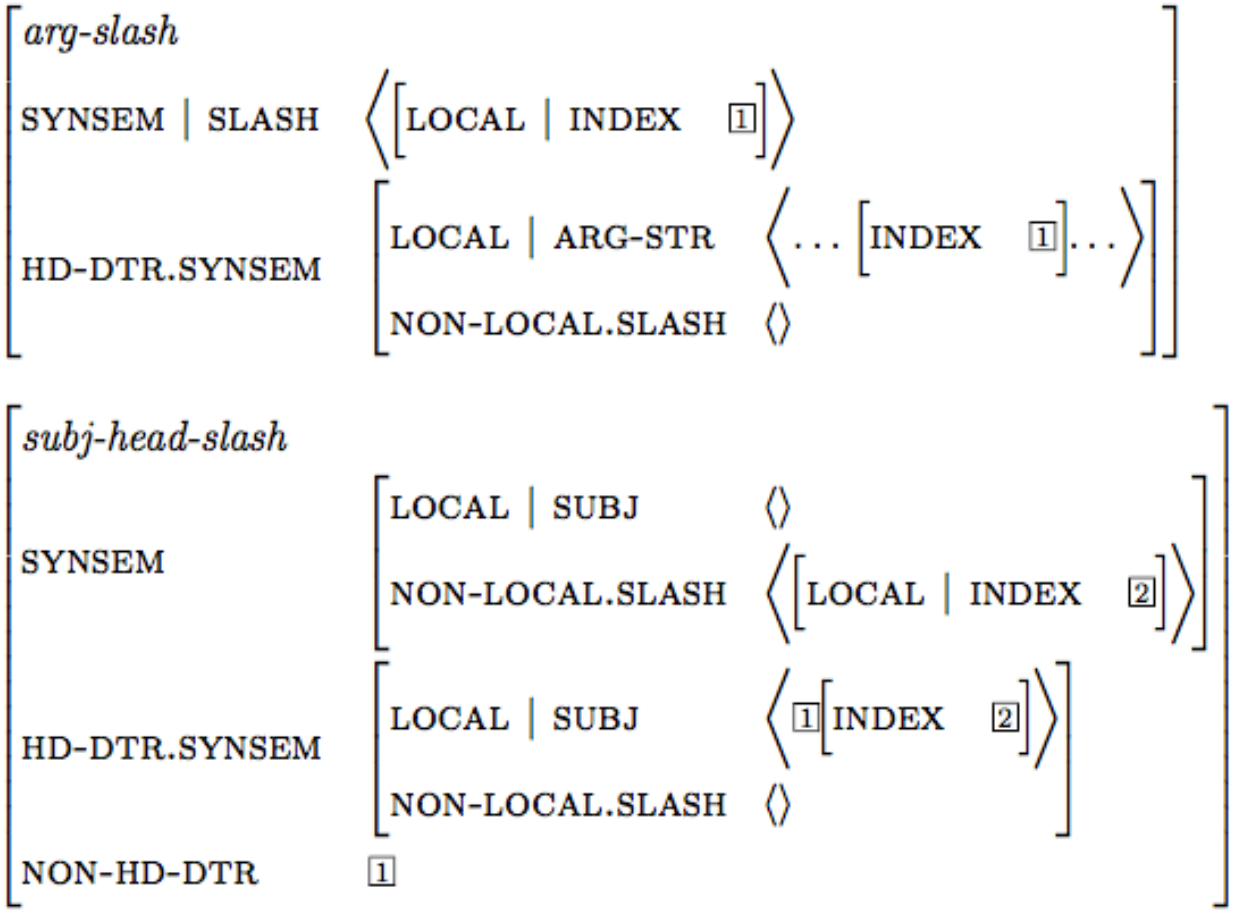
Resumption

- Coreference between external and local elements
  - Looks like resumption
  - Again, solution to resumption exists (see esp Crysmann 2012)
  - In addition to normal valence canceling, rule that moves valence item to slash

Resumption


subj-head
head-comp
arg-slash
head-opt-comp
head-opt-subj
subj-head-slash
head-comp-slash
head-opt-comp-slash
head-opt-subj-slash

Resumption


arg-slash
SYNSEM | SLASH ⟨[LOCAL | INDEX 1]⟩
HD-DTR.SYNSEM [LOCAL | ARG-STR ⟨... [INDEX 1] ...⟩
NON-LOCAL.SLASH ⟨⟩
subj-head-slash
SYNSEM [LOCAL | SUBJ ⟨⟩
NON-LOCAL.SLASH ⟨[LOCAL | INDEX 2]⟩
HD-DTR.SYNSEM [LOCAL | SUBJ ⟨1 [INDEX 2]⟩
NON-LOCAL.SLASH ⟨⟩
NON-HD-DTR 1

Resumption

- Free pronouns specify ICONS-KEY *focus-or-topic*
  - Since pronouns must agree with IS if present
- Syntactic PSRs give particular IS value
  - Left-clause dislocation: focus
  - Right-clause dislocaion: topic
  - These are guesses, based on frequencies and narrative structure of text, IS is difficult to glean from a corpus study

Conclusions

- DELPH-IN work has built up a lot of clever analyses, including argument optionality
- Phenomena can be grouped together by their properties (e.g., valence reduction, resumption)
- Boutique phenomena may be reducible to understood phenomena
  - We can demonstrate the usefulness of DELPH-IN style linguistic analysis
  - Toolsets can be more readily expanded to new languages (e.g., Grammar Matrix)

Conclusions

- Also I found some cool syntactic phenomenon it would be hard to find without a corpus study
  - Use of “temporal pronouns” (more tomorrow)
  - Free pronouns appear to require coindexing with syntactically-marked information structure (if available)
  - Zero-copula *only* (obligatorily?) with free pronouns

Future Directions

- Pretty much done* but…
  - A partial implementation would be good
  - Publication would be better
  - Plans to refine existing paper, submit (somewhere) alongside implementation covering at least sentences in paper

*Nuu-chah-nulth (Nootka) is taking up my language time these days